GLM-OCR Benchmark Report

Auto-generated by `tests/benchmark/run_benchmark.py`. Each row below is a real OCR run on this machine.

Environment

- **Device:** `mps` (`torch.float16`)
- **Model:** `zai-org/GLM-OCR`
- **Max long edge:** 1280 px
- **Model load time:** 17.3s (one-time at startup)

Summary

- Tested: **8** inputs
- Succeeded: **8** | Failed: **0**
- Total OCR time (excluding model load): **359.1s**
- Average per image: **44.9s**

Results table

#	Test	Category	Image size	Time	Output chars
1	Code screenshot	Code from a screen capture (like VS Code)	540×666	19.8s	1887
2	Scientific paper page	Research paper / academic PDF	1700×2200	100.6s	4750
3	Document page	Standard printed/PDF page baseline	843×596	61.2s	1147
4	Table	Spreadsheet / printed table	1080×1080	22.0s	402
5	Handwritten	Notes, whiteboard, sticky notes	1920×1653	24.8s	361
6	Seal / stamp	Specialty: stamped documents	399×314	2.2s	56
7	PDF (page 1 of 11)	Multi-page PDF processing	1700×2200	39.8s	2011
8	PDF (page 2 of 11)	Multi-page PDF processing	1700×2200	88.9s	4883

Detailed outputs

1. Code screenshot

- **Source:** `examples/source/code.png`
- **Real-world category:** Code from a screen capture (like VS Code)
- **What we're checking:** Preserves indentation, symbols, line breaks.
- **Prompt sent to model:** `Text Recognition:`
- **Image size:** 540×666
- **OCR time:** 19.76s
- **Output length:** 1887 characters

Output:

~~~
relationship between the beans. The only difference from previous exercises is the change
in the JNDI name element tag for the Address home interface:

```html
<local-jndi-name>AddressHomeLocal</local-jndi-name>
~~~

Because the Home interface for the Address is local, the tag is `<local-jndi-name>` rather than `<jndi-name>`.

The weblogic-cmp-rdbms-jar.xml descriptor file contains a number of new sections and elements in this exercise. A detailed examination of the relationship elements will wait until the next exercise, but there are some other changes to observe and examine.

The file contains a section mapping the Address `<cmp-field>` attributes from the `ejb-jar.xml` file in the database columns in the ADDRESS table, in addition to a new section related to the automatic key generation used for primary key values in this bean:

```
<weblogic-rdbms-bean>
  <ejb-name>AddressJB</ejb-name>
  <data-source-name>titan-dataSource</data-source-name>
  <table-name>ADDRESS</table-name>
  <field-map>
    <cmp-field>id</cmp-field>
    <dbms-column>ID</dbms-column>
  </field-map>
  <field-map>
    <cmp-field>street</cmp-field>
    <dbms-column>STREET</dbms-column>
  </field-map>
  <field-map>
    <cmp-field>city</cmp-field>
    <dbms-column>CITY</dbms-column>
  </field-map>
  <field-map>
    <cmp-field>state</cmp-field>
    <dbms-column>STATE</dbms-column>
  </field-map>
  <field-map>
    <cmp-field>zip</cmp-field>
    <dbms-column>ZIP</dbms-column>
  </field-map>
  <!-- Automatically generate the value of ID in the database on inserts using sequence
table -->
  <automatic-key-generation>
    <generator-type>NAMED_SEQUENCE_TABLE</generator-type>
    <generator-name>ADDRESS_SEQUENCE</generator-name>
    <key-cache-size>10</key-cache-size>
  </automatic-key-generation>
</weblogic-rdbms-bean>
```



```


### 2. Scientific paper page

- **Source:** `examples/source/paper.png`
- **Real-world category:** Research paper / academic PDF
- **What we're checking:** Multi-column scientific layout, equations, references.
```

```
- **Prompt sent to model:** `Text Recognition:`
- **Image size:** 1700×2200
- **OCR time:** 100.57s
- **Output length:** 4750 characters

**Output:**
```

B-5 First solution: Put $Q = x_1^2 + \cdots + x_n^2$. Since Q is homogeneous, P is divisible by Q if and only if each of the homogeneous components of P is divisible by Q. It is thus sufficient to solve the problem in case P itself is homogeneous, say of degree d.

Suppose that we have a factorization $P = Q^m R$ for some $m > 0$, where R is homogeneous of degree d and not divisible by Q; note that the homogeneity implies that

$$\sum_{i=1}^{n} x_i \frac{\partial R}{\partial x_i} = dR.$$

Write ∇^2 as shorthand for $\frac{\partial^2}{\partial x_1^2} + \cdots + \frac{\partial^2}{\partial x_n^2}$; then

$$0 = \nabla^2 P$$

$$= 2mn Q^{n-1} R + Q^m \nabla^2 R + 2 \sum_{i=1}^{n} 2mx_i Q^{m-1} \frac{\partial R}{\partial x_i}$$

$$= Q^m \nabla^2 R + (2mn + 4md) Q^{n-1} R.$$

Since $m > 0$, this forces R to be divisible by Q, contradiction.

Second solution: (by Noam Elkies) Retain notation as in the first solution. Let P_d be the set of homogeneous polynomials of degree d, and let H_d be the subset of P_d of polynomials killed by ∇^2, which has dimension $\geq \dim(P_d) - \dim(P_{d-2})$; the given problem amounts to showing that this inequality is actually an equality.

Consider the operator $Q \nabla^2$ (i.e., apply ∇^2 then multiply by Q) on P_d; its zero eigenspace is precisely H_d. By the calculation from the first solution, if $R \in P_d$, then

$$\nabla^2(QR) - Q \nabla^2 R = (2n + 4d)R.$$

Consequently, $Q^j H_{d-2j}$ is contained in the eigenspace of $Q \nabla^2$ on P_d of eigenvalue

$$(2n + 4(d - 2j)) + \cdots + (2n + 4(d - 2)).$$

In particular, the $Q^j H_{d-2j}$ lie in distinct eigenspaces, so are linearly independent within P_d. But by dimension counting, their total dimension is at least that of P_d. Hence they exhaust P_d, and the zero eigenspace cannot have dimension greater than $\dim(P_d) - \dim(P_{d-2})$, as desired.

Third solution: (by Richard Stanley) Write $x = (x_1, \ldots, x_n)$ and $\nabla = \left(\frac{\partial}{\partial x_1}, \ldots, \frac{\partial}{\partial x_n} \right)$. Suppose that $P(x) = Q(x)(x_1^2 + \cdots + x_n^2)$. Then

$$P(\nabla)P(x) = Q(\nabla)(\nabla^2)P(x) = 0.$$

On the other hand, if $P(x) = \sum_{\alpha} c_{\alpha} x^\alpha$ (where $\alpha = (\alpha_1, \ldots, \alpha_n)$ and $x^\alpha = x_1^{a_1} \cdots x_n^{a_n}$), then the constant term of $P(\nabla)P(x)$ is seen to be $ \sum_{\alpha} c_{\alpha}^2$. Hence $c_{\alpha} = 0$ for all α.

Remarks: The first two solutions apply directly over any field of characteristic zero. (The result fails in characteristic $p > 0$ because we may take $P = (x_1^2 + \cdots + x_n^2)x_1^2 + \cdots + x_n^2$. The third solution can be extended to complex coefficients by replacing $P(\nabla)$ by its complex conjugate, and again the result may be deduced for any field of characteristic zero. Stanley also suggests Section 5 of the arXiv e-print math.CO/0502363 for some algebraic background for this problem.

B-6 First solution: Let I be the identity matrix, and let J_x be the matrix with x's on the diagonal and 1's elsewhere. Note that $J_x - (x - 1)I$, being the all 1's matrix, has rank 1 and trace n, so has $n - 1$ eigenvalues equal to 0 and one equal to n. Hence J_x has $n - 1$ eigenvalues equal to $x - 1$ and one equal to $x + n - 1$, implying

$$\det J_x = (x + n - 1)(x - 1)^{n-1}.$$

On the other hand, we may expand the determinant as a sum indexed by permutations, in which case we get

$$\det J_x = \sum_{\pi \in S_n} \operatorname{sgn}(\pi) x^{\nu(\pi)}.$$

Integrating both sides from 0 to 1 (and substituting $y = 1 - x$) yields

$$\sum_{\pi \in S_n} \frac{\operatorname{sgn}(\pi)}{\nu(\pi)} + 1 = \int_0^1 (x + n - 1)(x - 1)^{n-1} dx$$

$$= \int_0^1 (-1)^{n+1}(n - y)y^{n-1} dy$$

$$= (-1)^{n+1} \frac{n}{n+1},$$

as desired.

Second solution: We start by recalling a form of the principle of inclusion-exclusion: if f is a function on the power set of ${1, \ldots, n}$, th

... [truncated for report; full length above]

```
### 3. Document page

- **Source:** `examples/source/page.png`
- **Real-world category:** Standard printed/PDF page baseline
- **What we're checking:** Regular document text + structure.
- **Prompt sent to model:** `Text Recognition:`
- **Image size:** 843×596
- **OCR time:** 61.19s
- **Output length:** 1147 characters

**Output:**
```

7.4 胶缝

7.4.1 采用胶缝传力的全玻璃幕墙，其胶缝必须采用硅酮结构密封胶。

7.4.2 全玻璃墙胶缝承载力应符合下列要求：

1 与玻璃面板平齐或突出的玻璃肋： $$\frac{q l}{2 t_1} \leqslant f_1$$ (7.4.2-1)

2 后置或骑缝的玻璃肋： $$\frac{q l}{2 t_2} \leqslant f_1$$ (7.4.2-2)

式中 q ——垂直于玻璃面板的分荷载设计值 (N/mm^2)，抗震设计时应包含地震作用计算的分布荷载设计值； l ——两肋之间的玻璃面板跨度（mm）； t_1 ——胶缝宽度，取玻璃面板截面厚度（mm）； t_2 ——胶缝宽度，取玻璃面板截面厚度（mm）； f_1 ——硅酮结构密封胶在风荷载作用下的强度设计值，取 $0.2N/mm^2$。

3 胶缝厚度应符合本规范第 5.6.5 条的要求，并不应小于 $6mm$。

7.4.3 当胶缝宽度不满足本规范第 7.4.2 条第 1、2 款的要求时，可采取附加玻璃板条或不锈钢条等措施，加大胶缝宽度。

8 点支承玻璃幕墙结构设计

8.1 玻璃面板

8.1.1 四边形玻璃面板可采用四点支承，有依据时也可采用六点支承；三角形玻璃面板可采用三点支承。玻璃面板支承孔边与板边的距离不宜小于 $70mm$。

8.1.2 采用浮头式连接件的幕墙玻璃厚度不应小于 $6mm$；采用沉头式连接件的幕墙玻璃厚度不应小于 $8mm$。

安装连接件的夹层玻璃和中空玻璃，其单片厚度也应符合上述要求。

8.1.3 玻璃之间的空隙宽度不应小于 $10mm$，且应采用硅酮建筑密封胶缝。

8.1.4 点支承玻璃支承孔周边应进行可靠的密封。当点支承玻璃为中空玻璃时，其支承孔周边应采取多道密封措

施。

8.1.5 在垂直于幕墙平面的风荷载和地震作用下，四点支承玻璃面板的应力和挠度应符合下列规定：

1 最大应力标准值和最大挠度可按考虑几何非线性的有限元方法计算，也可按下列公式计算： $$\sigma_{vk} = \frac{6 m w_k b^2}{t^2} \eta$$ (8.1.5-1) $$\sigma_{Ek} = \frac{6 m q_{Ek} b^2}{t^2} \eta$$ (8.1.5-2) $$d_t = \frac{p w_k b^4}{D} \eta$$ (8.1.5-3) $$\theta = \frac{w_k b^4}{E t^4} \text{ 或 } \theta = \frac{(w_k + 0.5 q_{Ek}) b^4}{E t^4}$$ (8.1.5-4)

式中 θ ——参数；

```
### 4. Table

- **Source:** `examples/source/table.png`
- **Real-world category:** Spreadsheet / printed table
- **What we're checking:** Captures cells, rows, and columns.
- **Prompt sent to model:** `Text Recognition:`
- **Image size:** 1080×1080
- **OCR time:** 21.99s
- **Output length:** 402 characters

**Output:**
```

项目 本期金额 上期金额 说明 非流动性资产处置损益，包括已计提资产减值准备的冲销部分 -342,273.79 -828,932.00 计入当期损益的政府补助，但与公司正常经营业务密切相关、符合国家政策规定、按照确定的标准享有、对公司损益产生持续影响的政府补助除外 10,102,396.61 6,417,015.44 除同公司正常经营业务相关的有效套期保值业务外，非金融企业持有金融资产和金融负债产生的公允价值变动损益以及处置金融资产和金融负债产生的损益 -1,132,468.57 -643,584.12 计入当期损益的对非金融企业收取的资金占用费 委托他人投资或管理资产的损益 对外委托贷款取得的损益 因不可抗力因素，如遭受自然灾害而产生的各项资产损失 单独进行减值测试的应收款项减值准备转回 企业取得子公司、联营企业及合营企业的投资成本小于取得投资时应享有被投资单位可辨认净资产公允价值产生的收益

```
### 5. Handwritten

- **Source:** `examples/source/handwritten.png`
- **Real-world category:** Notes, whiteboard, sticky notes
- **What we're checking:** How well does it handle handwriting.
- **Prompt sent to model:** `Text Recognition:`
- **Image size:** 1920×1653
- **OCR time:** 24.77s
- **Output length:** 361 characters

**Output:**
```

“双向的奔赴 才值得”

人这一辈子，总是在等，等下次，等不忙，等有钱，等有时间， 可等的没了选择，就等来了遗憾。人间的饭，吃一碗少一碗， 身边的人，见一面少一面，脚下的路走一步就少一步，其实，人生 就是一个减法，来日并不方长。

我很喜欢你，但是你不喜欢我了，你把我对你的喜欢消耗 完了，就到了该说再见的时候了。当然会有不舍呀，毕竟我知 道我们曾经是真的爱过对方，真真切切的，轰轰烈烈的，不 然不会在一起九年的时间，可我也不会遗憾，因为我觉得我 该付出的，该努力的都努力了，但是结果不好我认了。

人生其实没有走错路的地方，只是角度不一样。我们总会在 错的时间遇到对的人，也需要一些人用她的离开，用

你们这一 生的错过和再也得不到，教会你成长。我始终相信，付出真心 没有错，真心对一个人好没有错，信任一个人也没有错。你不爱

```
### 6. Seal / stamp

- **Source:** `examples/source/seal.png`
- **Real-world category:** Specialty: stamped documents
- **What we're checking:** Circular text in a stamp / seal.
- **Prompt sent to model:** `Text Recognition:`
- **Image size:** 399×314
- **OCR time:** 2.17s
- **Output length:** 56 characters

**Output:**
```

中国石油天然气股份有限公司四川销售分公司 510105711890927 发票专用章 5101055004432

```
### 7. PDF (page 1 of 11)

- **Source:** `examples/source/GLM-4.5V.pdf (page 1)`
- **Real-world category:** Multi-page PDF processing
- **What we're checking:** PDF pipeline: render page 1 -> OCR.
- **Prompt sent to model:** `Text Recognition:`
- **Image size:** 1700×2200
- **OCR time:** 39.79s
- **Output length:** 2011 characters

**Output:**
```

GLM-4.5V and GLM-4.1V-Thinking: Towards Versatile Multimodal Reasoning with Scalable Reinforcement Learning

GLM-V Team Zhipu AI & Tsinghua University (For the complete list of authors, please refer to the Contribution section)

Abstract

We present GLM-4.1V-Thinking and GLM-4.5V, a family of vision-language models (VLMs) designed to advance general-purpose multimodal understanding and reasoning. In this report, we share our key findings in the development of the reasoning-centric training framework. We first develop a capable vision foundation model with significant potential through large-scale pre-training, which arguably sets the upper bound for the final performance. We then propose Reinforcement Learning with Curriculum Sampling (RLCS) to unlock the full potential of the model, leading to comprehensive capability enhancement across a diverse range of tasks, including STEM problem solving, video understanding, content recognition, coding, grounding, GUI-based agents, and long document interpretation. In a comprehensive evaluation across 42 public benchmarks, GLM-4.5V achieves state-of-the-art performance on nearly all tasks among open-source models of similar size, and demonstrates competitive or even superior results compared to closed-source models such as Gemini-2.5-Flash on challenging tasks including Coding and GUI Agents. Meanwhile, the smaller GLM-4.1V-9B-Thinking remains highly competitive—achieving superior results to the much larger Qwen2.5-VL-72B on 29 benchmarks. We open-source both GLM-4.1V-9B-Thinking and GLM-4.5V. Code, models and more information are released at https://github.com/zai-org/GLM-V.

Figure 1: (A) GLM-4.5V achieves efficient scaling based on its compact predecessor, GLM-4.1V-9B-Thinking,

and compares favorably with Gemini-2.5-Flash, according to benchmark assessments. Table 2 presents full performance comparison. (B) Reinforcement learning substantially boosts the model's performance, with gains of up to +10.6% when experimented with GLM-4.5V.

```
### 8. PDF (page 2 of 11)

- **Source:** `examples/source/GLM-4.5V.pdf (page 2)`
- **Real-world category:** Multi-page PDF processing
- **What we're checking:** Same PDF, different page — confirms each page is independent.
- **Prompt sent to model:** `Text Recognition:`
- **Image size:** 1700×2200
- **OCR time:** 88.89s
- **Output length:** 4883 characters

**Output:**
```

1 Introduction

Vision-language models (VLMs) have become a crucial cornerstone of modern intelligent systems, enabling the perception and understanding of visual information beyond text. Over the past decade, as the intelligence level of models has advanced dramatically [36; 47; 20; 4], the complexity of corresponding multimodal intelligence tasks has increased accordingly. From solving scientific problems [67; 32; 34] to developing autonomous agents [21; 62; 43], the demands on VLMs have far surpassed simple visual content perception [31], with an increasing emphasis on advanced reasoning abilities. Recently, numerous studies have shown that long-form reasoning [60] and scalable reinforcement learning [42] can significantly enhance the ability of large language models (LLMs) to solve complex problems [23; 17]. Several previous works have attempted to enhance the reasoning capabilities of VLMs using similar paradigms [46; 33], but they mainly focus on specific domains. The open-source community currently also lacks a multimodal reasoning model that consistently outperforms traditional non-thinking models of comparable parameter scale across a broad range of scenarios and tasks.

In this report, we share our key findings in the development of GLM-4.1V-Thinking and GLM-4.5V, a family of VLMs designed to advance general-purpose multimodal reasoning. Our training framework is structured around a unified objective: to comprehensively enhance the model's reasoning capabilities through scalable reinforcement learning. For pre-training, we curate a broad and diverse corpus of knowledge-intensive multimodal data to equip the model with strong foundational capabilities, including (a) massive image-text pairs with accurate factual knowledge; (b) a self-curated academic corpus with interleaved image and text; (c) annotated documents and diagrams, instructional videos, and grounding data spanning both natural and synthetic images. This foundation model serves as a high-potential multimodal reasoning base for subsequent reinforcement learning. In the supervised fine-tuning phase, we construct carefully designed, domain-specific datasets that teach the model to perform effective reasoning with a standardized format across a wide range of tasks. Finally, we introduce Reinforcement Learning with Curriculum Sampling (RLCS) to drive large-scale, cross-domain reasoning capabilities. RLCS is a multi-domain reinforcement learning framework that combines curriculum learning with difficulty-aware sampling to improve training efficiency by selecting tasks and samples suited to the model's current competence. Our reinforcement learning process enhances training effectiveness and stability, and systematically improves the model's reasoning abilities through interaction and feedback across diverse domains.

To advance research in this field, we open-source GLM-4.1V-9B-Thinking (9 billion parameters) and GLM-4.5V (106B-A12B: 106 billion total parameters, 12 billion activated parameters), both of which achieve state-of-the-art performance among models of comparable size. In a comprehensive evaluation across 42 public benchmarks, GLM-4.5V achieves state-of-the-art performance on nearly all tasks, consistently outperforming strong open-source models such as Step-3 (321B-A38B) and Qwen-2.5-VL-72B, and achieves comparable or even superior performance on 22 benchmarks relative to the closed-source Gemini-2.5-Flash. Notably, GLM-4.5V advances the state-of-the-art for open-source VLMs of comparable size by roughly 10% or more across a wide range of tasks, including general VQA (MMStar, GeoBench), STEM (MMMU Pro, MathVerse, WeMath), chart understanding (ChartQAPro, ChartMuseum), long document understanding (MMLongBench-Doc), visual grounding (TreeBench, Ref-L4-test), spatial reasoning (ERQA), GUI agents (OSWorld, AndroidWorld, WebVoyagerSom, WebQuest), VLM coding (Design2Code, Flame-React-Eval), and video understanding (VideoMMMU, LVBench, MotionBench). GLM-4.1V-

... [truncated for report; full length above] ```

What was NOT tested (needs real-world input from you)

These categories from the original test list need physical or personal items I can't generate. To test them, just drop the image into the webpage:

Category	What to test
Book page	Phone photo of a real paperback page.
Shop receipt	Photo of any paper receipt.
Restaurant menu	Photo of a menu.
Magazine / newspaper	Multi-column print article.
Sticky note	Handwritten Post-it.
Whiteboard photo	Meeting whiteboard.
Street sign / shop signboard	Outdoor signage.
Medicine / nutrition label	Curved bottle or packet.
Business card	Designed small print.
License plate	Short string, high-stakes digit accuracy.
Angled / perspective photo	Document shot from 30° off-axis.
Low-light / glare photo	Reflective surface lighting.
Non-Latin script	Chinese, Hindi, Arabic, etc.
QR / barcode page	Text near a barcode.
Form with checkboxes	Boxes ticked vs un-ticked.
Passport / ID	Mixed photo + MRZ + fields. Your own only.
Old yellowed document	Tests contrast handling.